Geometry I

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October 16, 2026

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1 Geometry of curves

Definition 1.0.1. Let $I = (\alpha, \beta) \subseteq \mathbb{R}$ be an open interval. A C^k parametrized curve in $\mathbb{R}^n$ is a C^k map $\gamma : I \to \mathbb{R}^n, t \to \gamma(t) = (\gamma_1(t), \gamma_2(t), \ldots, \gamma_n(t))$.

In the following, we usually assume $\gamma \in C^\infty$.

Example 1.0.2.

$$\text{A unit circle} = \left\{(x, y) \mid x^2 + y^2 = 1\right\} = \{(\cos t, \sin t) \mid t \in \mathbb{R}\} = \{(\cos 2t, \sin 2t) \mid t \in \mathbb{R}\}.$$

A unit circle has two different parametrizations.

Definition 1.0.3. The velocity of a parametrized curve is $\gamma'(t)$ and its speed is $\|\gamma'(t)\|$. The curve is regular if $\gamma'(t) \neq 0$ for all $t \in I$.

Example 1.0.4. Consider the curve $\gamma(t) = (t^2, t^3)$. γ is a smooth curve, and

$$\gamma'(t) = (2t, 3t^2), \quad \|\gamma'(t)\| = \sqrt{4t^2 + 9t^4} = \sqrt{t^2 \underbrace{(4 + 9t^2)}_{>0}}.$$

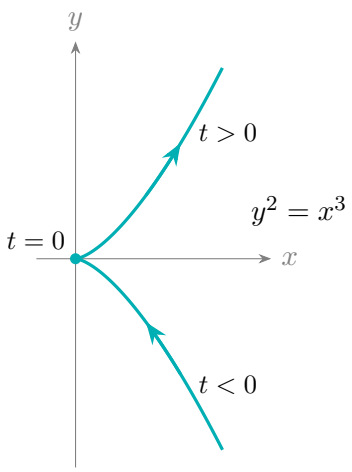
Also,

$$\gamma'(t) = 0 \Leftrightarrow \|\gamma'(t)\| = 0 \Leftrightarrow t = 0,$$

and the curve is not regular at $t = 0$.

If we paramatrize this curve, we know

$$\gamma(t) = (t^2, t^3) \Rightarrow \gamma(t) \in \left\{(x, y) \mid y^2 = x^3\right\}.$$



The cusp at the origin, where $\gamma'(0) = 0$. Arrows indicate increasing t.

Definition 1.0.5. Let $I, J \subseteq \mathbb{R}$ be open intervals. A regular change of parameter is a smooth bijection $\phi : J \to I$ with smooth inverse and $\phi'(u) \neq 0$ on J. The curve $\widetilde{\gamma} = \gamma \circ \phi$ is a reparametrization of γ.

Remark 1.0.6. If γ is regular then $\widetilde{\gamma}$ is also regular since

$$\widetilde{\gamma}'(u) = \gamma'(\phi(u))\phi'(u) \neq 0.$$

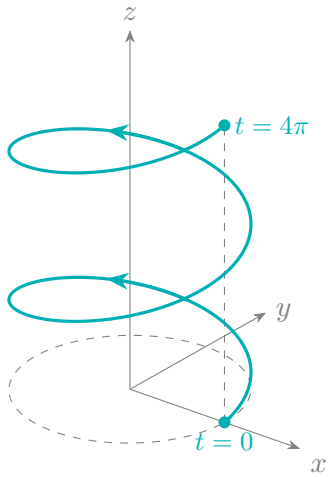
Definition 1.0.7 (Arc length)**.** For a C^1 curve $\gamma : [a, b] \to \mathbb{R}^n$, its length is

$$L(\gamma) = \int_a^b \|\gamma'(t)\| \, \mathrm{d}t.$$

Proposition 1.0.8

Let $\widetilde{\gamma} = \gamma \circ \phi$, where $\phi : [c, d] \to [a, b]$ is a regular change of parameter. Then $L(\widetilde{\gamma}) = L(\gamma)$.

Example 1.0.9. For $\gamma(t) = (a\cos t, a\sin t, bt)$, where $0 \le t \le T$ and $(a, b) \neq (0, 0)$ and $a, b > 0$. Then γ is called a helix.



A helix with $a = 1$, $b = \frac{1}{4}$, and $T = 4\pi$. Arrows indicate increasing t; the dashed circle is its projection onto the xy-plane.

We know

$$\begin{aligned} \gamma'(t) &= (-a\sin t, a\cos t, b) \\ \|\gamma'(t)\| &= \sqrt{a^2\sin^2 t + a^2\cos^2 t + b^2} = \sqrt{a^2 + b^2}. \\ \int_0^T \|\gamma'(t)\|\,\mathrm{d}t &= \sqrt{a^2 + b^2}T. \end{aligned}$$

Now suppose γ is regular. Define the arc-length function measured from t_0 is

$$s(t) = \int_{t_0}^{t} \|\gamma'(u)\|\,\mathrm{d}u, \quad \frac{\mathrm{d}s}{\mathrm{d}t} = \|\gamma'(t)\| > 0.$$

Proposition 1.0.10

Every regular curve on an open interval admits an orientation preserving parametrization by arc-length $\left\|\frac{\mathrm{d}\widetilde{\gamma}}{\mathrm{d}s}\right\| = 1$ where $\widetilde{\gamma} = \gamma(t(s))$.

Proof. Suppose $\gamma : (a, b) \to \mathbb{R}^n$ is regular, and $t_0 \in (a, b)$, and we have

$$\frac{\mathrm{d}s}{\mathrm{d}t} = \|\gamma'(t)\| > 0 \text{ on } (a, b).$$

Assume γ is restriction of a smooth curve, so t is a function of s. Thus

$$1 = \frac{\mathrm{d}s}{\mathrm{d}s} = \frac{\mathrm{d}s}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}s} = \|\gamma'(t)\|\frac{\mathrm{d}t}{\mathrm{d}s} \Rightarrow \frac{\mathrm{d}t}{\mathrm{d}s} = \frac{1}{\|\gamma'(t)\|}.$$

Hence,

$$\widetilde{\gamma}(s) = \gamma(t(s)),$$

and by chain rule

$$\frac{\mathrm{d}\widetilde{\gamma}}{\mathrm{d}s} = \frac{\mathrm{d}\gamma}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}s} = \gamma'(t)\cdot\frac{1}{\|\gamma'(t)\|} \Rightarrow \left\|\frac{\mathrm{d}\widetilde{\gamma}}{\mathrm{d}s}\right\| = 1.$$

■

Suppose γ is a regular C^2 curve. After reparametrizing by arc length, write $\gamma : J \to \mathbb{R}^n$, where J is an open interval with parameter s, and $\|\gamma'(s)\| = 1$. For each $s \in J$, the derivative $\gamma'(s)$ is a tangent vector to the curve at $\gamma(s)$, pointing in the direction of increasing s. Since it has length one, we call it the *unit tangent vector* and denote it by $T(s)$. Thus, T is the vector-valued C^1 function assigning to each parameter value its unit tangent vector:

$$T : J \to \mathbb{R}^n, \qquad T(s) = \gamma'(s) = (T_1(s), \ldots, T_n(s)),$$

where each $T_j : J \to \mathbb{R}$ is a scalar-valued C^1 function. The derivative of T is a vector, defined by a limit in $\mathbb{R}^n$ and computed componentwise:

$$T'(s) = \lim_{h\to 0}\frac{T(s+h)-T(s)}{h} = (T_1'(s), \ldots, T_n'(s)) = \gamma''(s).$$

For any C^1 vector-valued functions $U, V : J \to \mathbb{R}^n$, their Euclidean inner product is the scalar-valued function

$$s \longmapsto \langle U(s), V(s)\rangle = \sum_{j=1}^{n} U_j(s)V_j(s) \in \mathbb{R}.$$

Thus, differentiating the inner product means differentiating a finite sum of products of real-valued functions. Applying the ordinary product rule to each summand gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}s}\langle U(s), V(s)\rangle &= \sum_{j=1}^{n}\big(U_j'(s)V_j(s) + U_j(s)V_j'(s)\big)\\ &= \langle U'(s), V(s)\rangle + \langle U(s), V'(s)\rangle.\end{aligned}$$

In particular, $\langle T(s), T(s)\rangle = \|T(s)\|^2 = 1$. Taking $U = V = T$ in the formula above, we obtain

$$\begin{aligned}0 &= \frac{\mathrm{d}}{\mathrm{d}s}\langle T(s), T(s)\rangle\\ &= \langle T'(s), T(s)\rangle + \langle T(s), T'(s)\rangle\\ &= 2\langle T'(s), T(s)\rangle,\end{aligned}$$

where the last equality uses the symmetry of the Euclidean inner product. Hence

$$\langle T'(s), T(s)\rangle = \langle \gamma''(s), \gamma'(s)\rangle = 0.$$

Thus, $T'(s)$ is orthogonal to the unit tangent $T(s)$. This includes the possibility $T'(s) = 0$. Since T has constant length, its derivative describes only a change in direction. The magnitude of this change per unit arc length is the curvature, defined below.

Definition 1.0.11. For a unit speed C^2 regular curve, the curvature is

$$\kappa(s) = \|T'(s)\| = \|\gamma''(s)\|.$$

Remark 1.0.12. If $\gamma(t)$ is regular, then

$$\kappa(t) = \left\|\frac{\mathrm{d}T}{\mathrm{d}s}\right\| = \left\|\frac{\mathrm{d}T}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}s}\right\| = \frac{\left\|\frac{\mathrm{d}T}{\mathrm{d}t}\right\|}{\|\gamma'(s)\|}$$

Proposition 1.0.13

Let $\gamma : I \to \mathbb{R}^3$ be a regular C^2 curve. Then

$$\kappa = \frac{\sqrt{\|\gamma'\|\|\gamma''\|^2 - \langle \gamma', \gamma'' \rangle^2}}{\|\gamma'(t)\|^3},$$

and when $n = 3$, we have

$$\kappa = \frac{\|\gamma' \times \gamma''\|}{\|\gamma\|^3}.$$